

DOCUMENT NAME

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1. Introduction

The Database Design maps the logical data model for the Elderly Health Monitoring and Alert System to the target database management system, with specific consideration to the system's real-time performance, data integrity, and alert efficiency. The Database Design converts logical or conceptual data constructs (e.g., patient records, vital signs, alert rules) to physical data constructs (e.g., tables, indexes, stored procedures) of the target Database Management System.

1.1 Purpose

The purpose of the Database Design is to ensure that every database transaction, especially those involving the collection of real-time vital signs (heart rate, SpO2, temperature) and the generation of urgent alerts, meets or exceeds the system's performance and reliability requirements. This document takes into account the volume of continuous sensor data, the need for quick data retrieval for monitoring dashboards, and the strict latency requirements for the alert mechanism to produce a robust schema and environment that will meet necessary performance and ensure timely care.

1.2 Scope

The Database Design Document for the Elderly Health Monitoring and Alert System has the following key objectives:

- To describe the design of the database, that is, a collection of related health and profile data (e.g., Elderly Profile, Health Metrics, Alert History, Device Status) stored in one or more computerized files that can be accessed by caregivers, medical staff, or the system's alert services via a DBMS.
- To serve as a basis for implementing the database and related software units (e.g., the Data Ingestion Module, the Alert Engine, and the Caregiver Dashboard). It provides the software development team visibility into the data architecture and provides essential information necessary for the development of real-time monitoring and alert functionalities.

1.3 Introduction about MySQL

MySQL is an open-source Relational Database Management System (RDBMS) that uses Structured Query Language (SQL) to manage and manipulate data. It is one of the most popular databases in the world, developed and maintained by Oracle Corporation. MySQL allows users to store, retrieve, and manage structured data in the form of tables, using relationships between them to ensure data integrity.

MySQL is known for its high performance, reliability, and scalability, making it suitable for both small-scale and enterprise-level applications. It supports multiple operating systems, including Windows, Linux, and macOS, and can be integrated easily with web technologies such as PHP, Python, and Node.js.

In a relational database like MySQL, data is stored in tables (relations) that consist of rows (records) and columns (fields). Each table represents an entity, and relationships are defined through primary keys and foreign keys.

2. Database Diagram

2.1. Table Overview

Table Name	Short Description
Users	This table stores information of caregivers/relatives who manage elderly people and register devices.
Admin	This table stores system administrator accounts who review and approve device registrations.
Elderly_Users	This table stores detailed information of elderly people being monitored by the system.
Devices	This table stores device information such as type, owner, and status of each monitoring device.
Device_Registrations	This table stores device registration requests made by users and reviewed by admins.
Device_assignments	This table stores the assignment history of devices to users, including time and location.
Weight and BMI	This table stores the weight, height, and calculated BMI records for the elderly users.
Device_Assignments	Tracks the devices currently assigned to users, including device status and assignment metadata.
Fall Events	Stores all fall detection events captured by AI analysis and Sensor, including timestamps, severity, and related user information.
Doctor_ai_chat_history	Stores interactions between users and the AI doctor assistant, including questions, responses, detected conditions, and timestamps.
Water_intake_records	Stores records of users' daily water consumption, including intake time and volume for hydration tracking.

2.1 Table Relationship Diagram

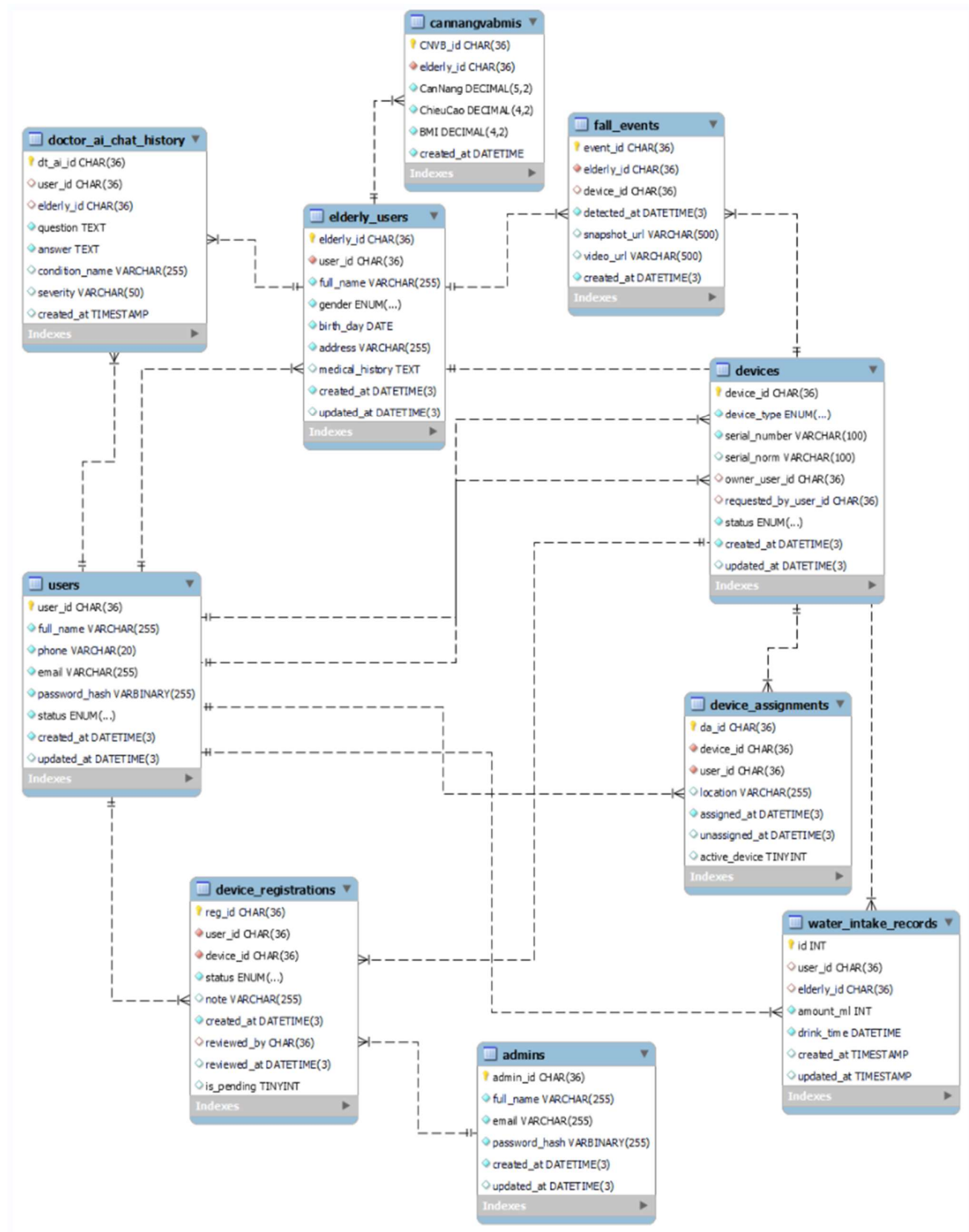


Figure 1.: Table Relationship Diagram

3. Database Design for Sprint

3.1. Table Users

Users				
Id	Field	Type	Constrain	Description
1	user_id	CHAR(36)	Primary Key, NOT NULL	UUID for each user
2	full_name	VARCHAR(255)	NOT NULL	The full name of the user.
3	phone	VARCHAR(20)	UNIQUE, NOT NULL	The phone number, must be unique
4	email	VARCHAR(255)	UNIQUE, NOT NULL	The email, must be unique
5	password_hash	VARBINARY(255)	NOT NULL	The encrypted password
6	status	ENUM(...)	NOT NULL, Default 'inactive'	The account status
7	created_at	DATETIME(3)	NOT NULL, DEFAULT CURRENT_TIMESTAMP(3)	Creation time, auto-assign
8	updated_at	DATETIME(3)	DEFAULT NULL, ON UPDATE CURRENT_TIMESTAMP(3)	Last update time, auto update

3.2. Table Admins

Admins				
Id	Field	Type	Constrain	Description
1	admin_id	CHAR(36)	Primary key, NOT NULL	UUID for each admin account.
2	full_name	VARCHAR(255)	NOT NULL	The full name of the admin.
3	email	VARCHAR(255)	UNIQUE, NOT NULL	Email used to login must be unique
4	password_hash	VARBINARY(255)	NOT NULL	Hashed password of the admin.
5	created_at	DATETIME(3)	NOT NULL, DEFAULT CURRENT_TIMESTAMP(3)	Account creation time, auto-assign.
6	updated_at	DATETIME(3)	DEFAULT NULL, ON UPDATE CURRENT_TIMESTAMP(3)	Last update time, auto update.

3.3. Table Elderly_Users

Elderly_Users				
Id	Field	Type	Constrain	Description
1	Elderly_id	CHAR(36)	Primary key, NOT NULL	UUID for each senior profile
2	user_id	CHAR(36)	Foreign Key, NOT NULL	Link profiles to Users accounts
3	full_name	VARCHAR(255)	NOT NULL	The full name of the elderly person
4	gender	ENUM(...)	NOT NULL	Gender limited to the values “male”, “female”, “other”.
5	birth_day	DATE	NOT NULL	The date of birth
6	address	VARCHAR(255)	NULL	Address (Included in Figure 1, not in Figure 2).
7	Created_at	DATETIME(3)	NOT NULL, DEFAULT CURRENT_TIMESTAMP(3)	Profile creation time
8	updated_at	DATETIME(3)	DEFAULT NULL, ON UPDATE CURRENT_TIME	Last update time.

3.4. Table Devices

Devices				
Id	Field	Type	Constrain	Description
1	device_id	CHAR(36)	Primary key, NOT NULL	A unique identifier (UUID) for each device.
2	device_type	ENUM('CAMERA', 'JEWELRY')	NOT NULL	Device type (Camera or Smart Jewelry).
3	serial_number	VARCHAR(100)	UNIQUE, NOT NULL	The official serial number of the device
4	serial_normalized	VARCHAR(100)	Generated Always	Normalized lowercase version of the serial number, used for comparison purposes
5	owner_user_id	CHAR(36)	Foreign Key → USERS(user_id). NULL allowed	The user currently owns the device. May be NULL if not owned.
6	requested_by_user_id	CHAR(36)	Foreign Key → USERS(user_id). NULL allowed	The user has requested to own or register the device. May be NULL
7	status	ENUM('active', 'inactive', 'pending', 'rejected')	NOT NULL, Default 'pending'	Current status of the device in the system

8	created_at	DATETIME(3)	Default CURRENT_TIMESTAMP	Date and time the device record was created
9	updated_at	DATETIME(3)	Auto Update	Date and time the device record was last updated.

3.5. Table Device_Registrations

Device Registrations				
Id	Field	Type	Constraint	Description
1	reg_id	CHAR(36)	Primary Key	A unique identifier (UUID) for each device registration request.
2	user_id	CHAR(36)	Foreign Key → Users(user_id), NOT NULL	The user who submits the device registration request.
3	device_id	CHAR(36)	Foreign Key → DEVICES(device_id), NOT NULL	The device being requested for registration.
4	device_type	ENUM('CAMERA', 'JEWELRY')	NOT NULL	The type of the device being registered.
5	status	ENUM('pending', 'approved', 'rejected')	NOT NULL, Default 'pending'	The current status of the registration request.
6	note	VARCHAR(255)	NULL allowed	Additional comments or reasons related to the request.
7	created_at	DATETIME(3)	Default CURRENT_TIMESTAMP	The date and time when the registration request was created.
8	reviewed_by	CHAR(36)	Foreign Key → ADMINS(admin_id), NULL allowed	The admin who reviews and approves/rejects the request.
9	reviewed_at	DATETIME(3)	NULL allowed	The date and time when the request was reviewed.

3.6. Table Device_Assignments

Device_Assignments				
Id	Field	Type	Constrain	Description
1	da_id	CHAR(36)	Primary key	A unique identifier (UUID) for each device assignment record.
2	device_id	CHAR(36)	Foreign Key → DEVICES(device_id), NOT NULL	The device that is assigned to the user.
3	user_id	CHAR(36)	Foreign Key → USERS(user_id), NOT NULL	The user who is assigned to use the device.
4	location	VARCHAR(255)	NULL allowed	The location where the device is used or installed.
5	assigned_at	DATETIME(3)	Default CURRENT_TIMESTAMP	The date and time when the device was assigned to the user.
6	unassigned_at	DATETIME(3)	NULL allowed	The date and time when the device was returned or unassigned.
7	active_device	TINYINT	Generated Always	Indicates whether the device is currently active (1) or not (NULL).

3.7. Table Weight and BMI

Weight and BMI				
Id	Field	Type	Constrain	Description
1	cnvb_id	VARCHAR(255)	Primary Key, NOT NULL	A unique identifier for the weight and BMI record.
2	elderly_id	CHAR(36)	Foreign Key → Elderly_Users(elderly_id), NOT NULL	The ID of the elderly user this record belongs to.
3	weight	DECIMAL(5, 2)	NOT NULL	The recorded weight (in kg).
4	height	DECIMAL(5, 2)	NOT NULL	The recorded height (in meters).
5	bmi	DECIMAL(5, 2)	Generated Always	The calculated Body Mass Index (BMI).
6	created_at	DATETIME	Default CURRENT_TIMESTAMP	The date and time the record was created.

3.8. Table Fall Events

FALL_EVENTS				
Id	Field	Type	Constrain	Description
1	event_id	CHAR(36)	Primary key	A unique identifier (UUID) for each fall event.
2	elderly_id	CHAR(36)	Foreign Key → ELDERLY_USERS(elderly_id), NOT NULL	The elderly person who experienced the fall.
3	device_id	CHAR(36)	Foreign Key → DEVICES(device_id), NULL allowed	The device (camera) that detected the fall
4	detected_at	DATETIME(3)	Default CURRENT_TIMESTAMP(3), NOT NULL	The exact time when the fall was detected.
5	snapshot_url	VARCHAR(500)	NULL allowed	The snapshot image URL captured at the fall moment.
6	video_url	VARCHAR(500)	NULL allowed	The video URL stored in cloud (MP4/HLS).
7	created_at	DATETIME(3)	Default CURRENT_TIMESTAMP(3), NOT NULL	The timestamp when this record was created.

3.9. Table doctor_ai_chat_history

Doctor_ai_chat_history				
Id	Field	Type	Constrain	Description
1	dt_ai_id	CHAR(36)	Primary key	A unique identifier (UUID) for each AI doctor chat record.
2	user_id	CHAR(36)	Foreign Key → users(user_id), NULL allowed	The caregiver or user who asked the question.
3	elderly_id	CHAR(36)	Foreign Key → elderly_users(elderly_id), NULL allowed	The elderly person related to the medical context.
4	question	TEXT	NOT NULL	The question asked by the user

				to the AI doctor.
5	answer	TEXT	NOT NULL	The AI-generated medical answer.
6	condition_name	VARCHAR(255)	NULL allowed	The medical condition identified (if any).
7	severity	VARCHAR(50)	NULL allowed	Severity level of the condition (mild, moderate, severe...).
8	created_at	TIMESTAMP	Default CURRENT_TIMESTAMP	The timestamp when the chat entry was created.
9	Idx_user_id	INDEX	INDEX (user_id)	Improves lookup for chats by user.
10	idx_elderly_id	INDEX	INDEX (elderly_id)	Improves lookup for chats by elderly person.
11	idx_created_at	INDEX	INDEX (created_at)	Improves sorting/searching by date.

3.10. Table water_intake_records

water_intake_records				
Id	Field	Type	Constrain	Description
1	id	INT	AUTO_INCREMENT, Primary Key	A unique identifier (UUID) for each device.
2	user_id	CHAR(36)	Foreign Key → users(user_id), NULL allowed	Device type (Camera or Smart Jewelry).
3	elderly_id	CHAR(36)	Foreign Key → elderly_users(elderly_id), NULL allowed	The official serial number of the device
4	amount_ml	INT	NOT NULL	Normalized lowercase version of the serial number, used for comparison purposes
5	drink_time	DATETIME	NOT NULL	The user currently owns the device. May be NULL if not owned.
6	created_at	TIMESTAMP	Default CURRENT_TIMESTAMP	The user has requested to own or register the device. May be NULL
7	updated_at	TIMESTAMP	Default CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP	Timestamp when the record was last updated.

4. Hardware and Software Requirements

This section provides an overview of hardware and software requirements. Below are descriptions of the technological components of the Elderly Health Monitoring and Alert System:

Attributes of Elderly Health Monitoring and Alert System	
Attributes	Descriptions
Database	MySQL
Software	Nodejs, React
Hardware	Computer
Library	